

Kaushik Nath Bhaumik

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Pharmaceutical and bio-chemistry researcher with a multidisciplinary skill set, including expertise in peptide chemistry, analytical chemistry, drug delivery, and biological assays.

Experience

Postdoctoral Research Fellow

Nov 2025 – Present

University of Brighton, Brighton, United Kingdom

Responsibilities:

- Designed peptide-based nanocarriers for targeted delivery of therapeutics for wound healing.
- Managed research planning, timelines, and deliverables within the EU Horizon Europe INJECTHEAL project in collaboration with European academic and industrial partners.
- Coordinated multidisciplinary teams and contributed to grant management, reporting, and stakeholder communication.
- Supervised and mentored students and junior researchers, providing technical and project guidance.
- Performed peptide synthesis using solid-phase peptide synthesis (SPPS) and conducted physicochemical characterisation.
- Applied biomaterials and nanocarrier characterisation techniques, including in vitro biocompatibility testing.
- Analysed and interpreted physicochemical and biological data to support publications and project deliverables.
- Ensured compliance with research governance, ethics, and health and safety standards while managing laboratory resources.

Lecturer in Drug Delivery & Pharmaceuticals (Part-time)

Feb 2025 – Jul 2025

University of East Anglia, Norwich, United Kingdom

Responsibilities:

- Teaching Pharmaceuticals (including the giving of lectures, seminars, tutorials and lab practicals) M. Pharm. and B. Sc. in Pharmacology and Drug Discovery degrees.
- Contributed to curriculum development through the design of teaching and learning materials and collaborative module organisation with academic colleagues.
- Assessing students according to prescribed specifications including marking examinations and providing written and oral feedback.
- Undertaking all administration and academic duties required to sustain the delivery of high-quality teaching.
- Supervision of undergraduate project students.
- Research activities focused on development of oral delivery strategies for biologics such as peptides and proteins.

PhD Researcher

Sep 2018 – Nov 2024

Doctoral School of Pharmaceutical Sciences, University of Szeged, Szeged, Hungary

Responsibilities:

- Synthesised antimicrobial peptides and peptidomimetic foldamers using manual and automated solid-phase peptide synthesis (SPPS).
- Purified and analysed peptides with HPLC-MS, CD spectroscopy, UV-Vis spectroscopy etc.
- Formulated lipid vesicles tailored for biophysical analysis and cell delivery.
- Developed biophysical assays, employing NMR and fluorescence spectroscopy, to analyse membrane activity of the peptides.
- Performed protein expression in *E. coli* and protein binding assays with ITC to investigate molecular interactions.
- Collaborated with biological team and provided assistance in conducting *in vitro* and *in vivo* assays such as MIC, FACS etc.
- Demonstrated proficiency in data analysis and visualisation using RStudio.

Senior Officer - Quality Control

Oct 2017 – Feb 2018

Sun Pharmaceutical Industries Ltd., Halol, India

Responsibilities:

- Performed analysis of laboratory samples (Finished APIs, Intermediates, Raw Materials, Recovered Solvent, In-process material, Working Standards) on HPLC as per approved Specification.
- Performed troubleshooting and ensured laboratory instruments are maintained and calibrated as per schedule.
- Maintained inventory of laboratory chemical and documents/spares in HPLC laboratory.
- Ensured compliance to cGxP and applicable safety standards in the QC laboratory.
- Maintained accurate records, including log books, for relevant analyses.

Junior Research Fellow

Dec 2015 – Oct 2017

Translational Health Science and Technology Institute, Department of Biotechnology, Govt. of India, Faridabad, India

Responsibilities:

- Formulated drug delivery systems such as lipid nanoparticles, biodegradable polymeric and hydrogel drug implants, lipid drug conjugates.

- Developed and validated HPLC methods for drug encapsulation and *in vitro/ in vivo* pharmacokinetic studies of the formulations.
- Conducted analysis using instruments such as NMR, DLS, DSC-TGA, UV-Vis Spectrophotometer, TFF systems, Dissolution test apparatus.
- Analysed drug content from blood/plasma, animal organs using HPLC for PK studies and performed cytokine assays.
- Performed cellular bio-distribution and trafficking in mammalian cell line using fluorescence and confocal microscopes.

Exam Invigilator & Administrator (variable hours)

May 2024 – Present

City College Norwich, Norwich, United Kingdom

Responsibilities:

- Invigilated paper-based and computer-based exams, ensuring compliance with regulations and maintaining security.
- Prepared exam rooms, distributed materials, and provided instructions, ensuring an organised process.
- Monitored students to prevent misconduct and promptly addressed issues.
- Acted as scribe and reader for students with special needs.

Education

PhD (Pharmaceutical Chemistry)

Sep 2018 – Nov 2024

Doctoral School of Pharmaceutical Sciences, University of Szeged, Szeged, Hungary

Supervisor: Prof. Tamás A. Martinek

PhD Thesis: Enhancing antibiotic efficacy through antimicrobial peptide-mediated modulation of bacterial electrophysiology: The sub-MIC approach.

Master of Pharmacy (Pharmaceutical Chemistry)

Jul 2013 – Jun 2015

Department of Pharmaceutical Chemistry, Tripura University, Tripura, India

Supervisor: Dr. Rajat Ghosh

Dissertation: Synthesis, characterization and biological evaluation of thiazine ring based newly synthesized heterocyclic compounds.

Bachelor of Pharmacy (Pharmaceutical Sciences)

Aug 2009 – Jun 2013

Regional Institute of Pharmaceutical Science & Technology

Tripura University, Tripura, India

Supervisor: Dr. Tejendra Bhakta

Dissertation: Evaluation of free-radical scavenging activity of different Fractioned Extracts of *Ichinocarpus frutescence*.

Skills

Peptide Synthesis

Manual and Automated Solid Phase Peptide Synthesis (SPPS).

Analytical Instrumentation

HPLC-MS, NMR, ITC, CD spectroscopy, DLS, FTIR, UV-VIS spectroscopy, DSC-TGA, Fluorescence spectroscopy.

Drug Delivery

Liposomes, Lipid nanoparticles (SLN & NLC), Polymeric microparticle, Polymeric drug implants, Hydrogel implants.

Biology

Antibacterial assays, Cell culture, Protein expression, Assay development and Confocal microscopy.

Data Analysis & Visualisation

Data visualization using RStudio.

Publications

- **Bhaumik KN**, Spohn R, Dunai A, Daruka L, Hetényi A, et al. Chemically diverse antimicrobial peptides induce hyperpolarization of the bacterial membrane. *Communication Biology*. 2024; 7, 1-12.
- **Bhaumik KN**, Hetényi A, Olajos G, Martins A, et al. Rationally designed foldameric adjuvants enhance antibiotic efficacy via promoting membrane hyperpolarization. *Molecular Systems Design & Engineering*. 2022;7(1):21-33.
- Hetényi A, Szabó E, Imre N, **Bhaumik KN**, et al. α/β -Peptides as Nanomolar Triggers of Lipid Raft-Mediated Endocytosis through GM1 Ganglioside Recognition. *Pharmaceutics*. 2022;14(3):580.
- Banerjee S, Roy S, **Bhaumik KN**, Pillai J. Mechanisms of the effectiveness of lipid nanoparticle formulations loaded with anti-tubercular drugs combinations toward overcoming drug bioavailability in tuberculosis. *Journal of Drug Targeting*. 2020;28(1):55-69.
- Banerjee S, Roy S, **Bhaumik KN**, Kshetrapal P, Pillai J. Comparative study of oral lipid nanoparticle formulations (LNFs) for chemical stabilization of antitubercular drugs: physicochemical and cellular evaluation. *Artificial Cells, Nanomedicine, and Biotechnology*. 2018;46(1):540-58.

Conferences

- **Bhaumik KN**, Hetényi A, Olajos G, Martins A, Spohn R, Németh L, et al. Novel foldameric peptide adjuvants counter antimicrobial resistance via promoting membrane hyperpolarization. 36th European Peptide Symposium and 12th International Peptide Symposium. *Barcelona, Spain 2022. August 28 - September 2*.
- **Bhaumik KN**, Hetényi A, Martinek TA, et al. Host defence peptides (HDPs) hyperpolarization membrane by selective ion transport at sub-MIC concentration. *Peptide Chemistry and Chemical Biology Symposium. Balatonszemes, Hungary 2022. May 30 - June 1*.

- **Bhaumik KN**, Hetényi A, Olajos G, Martins A, Spohn R, Németh L, et al. Rationally Designed Foldameric Adjuvants Enhance Antibiotic Efficacy via Promoting Membrane Hyperpolarization. *10th Interdisciplinary Doctoral Conference. Pécs, Hungary 2021. November 12-13.*
- **Bhaumik KN**, Hetényi A, Olajos G, Martins A, Spohn R, Németh L, et al. Rationally Designed Foldameric Adjuvants Enhance Antibiotic Efficacy via Promoting Membrane Hyperpolarization. MTA Peptidkémiai Munkabizottság Ülésére (MTA Peptide Chemistry Working Committee Meeting) *Balatonszemes, Hungary 2021. October 11-13.*

Awards

Stipendium Hungaricum Scholarship

Tempus Public Foundation

PhD Scholarship award for 4.5 years (June 2018).

European Peptide Society Symposium Grant

To attend the 36th European and 12th International Peptide Symposium, August 28 to September 2, 2022 at Sitges (Barcelona), Spain. (Apr 2022)